

APK Structure Overview – Android Package Kit

Reverse engineering mein sab se pehla aur important step APK ki structure ko samajhna hai. APK (Android Package Kit) ek Android installation file hai jo aam taur par ZIP format mein hoti hai. Jab aap kisi APK ko decompress karte hain (jaise unzip tool se ya apktool ke zariye), toh aapko uske andar kuch specific files aur folders milte hain jo app ke bare mein important information rakhte hain.

APK Ke Important Components

AndroidManifest.xml

- Yeh ek required configuration file hai jo app ke bare mein basic information deti hai
- Ismein package name, app components (activities, services, receivers), aur permissions list hoti hai
- Minimum Android version aur target SDK version bhi isi file mein define hota hai
- OWASP ke mutabiq, yeh reverse engineering ke liye sab se important source of information hai

- Ismein backup allowance, debuggable flag, aur exported components jaise security-relevant attributes bhi hote hain

classes.dex

- Yeh app ka main runnable file hai jo Dalvik Virtual Machine par execute hota hai
- Ismein app ki saari operating instructions aur runtime data hoti hain
- Agar app bohot bada hai toh multiple DEX files ho sakti hain (classes2.dex, classes3.dex, etc.)
- Yeh Java/Kotlin code ko Dalvik bytecode mein convert karke rakhti hai
- Reverse engineering ke dauran isi file ko decompile karke Java source code ya Smali code mein convert kiya jata hai

res/ Folder

- Ismein app ki saari resources (images, UI layouts, strings, colors, dimensions) store hoti hain
- Sub-folders different resource types aur configurations ke hisaab se organized hote hain:
 - drawable/ – Images aur icons
 - layout/ – UI layouts (XML files)
 - values/ – Strings, colors, dimensions, styles
 - mipmap/ – Launcher icons (different densities ke liye)

resources.arsc

- Ek compiled binary resource file hai jo code aur resources ke beech link provide karti hai
- Ismein resource IDs aur unke corresponding values store hote hain
- App runtime par resources ko access karta hai isi file ke zariye

lib/ Folder

- Native libraries (machine code) ko store karta hai
- Different CPU architectures ke hisaab se sub-folders mein organized hota hai:
 - armeabi-v7a – 32-bit ARM
 - arm64-v8a – 64-bit ARM
 - x86 – 32-bit Intel
 - x86_64 – 64-bit Intel
- Ismein .so files (shared libraries) hoti hain jo C/C++ code par based hain

assets/ Folder

- Raw files store karta hai jinhein Android resource system process nahi karta
- Ismein aam taur par custom fonts, audio files, configuration files, aur pre-compiled libraries hoti hain
- Yeh files AssetManager class ke zariye access ki jati hain

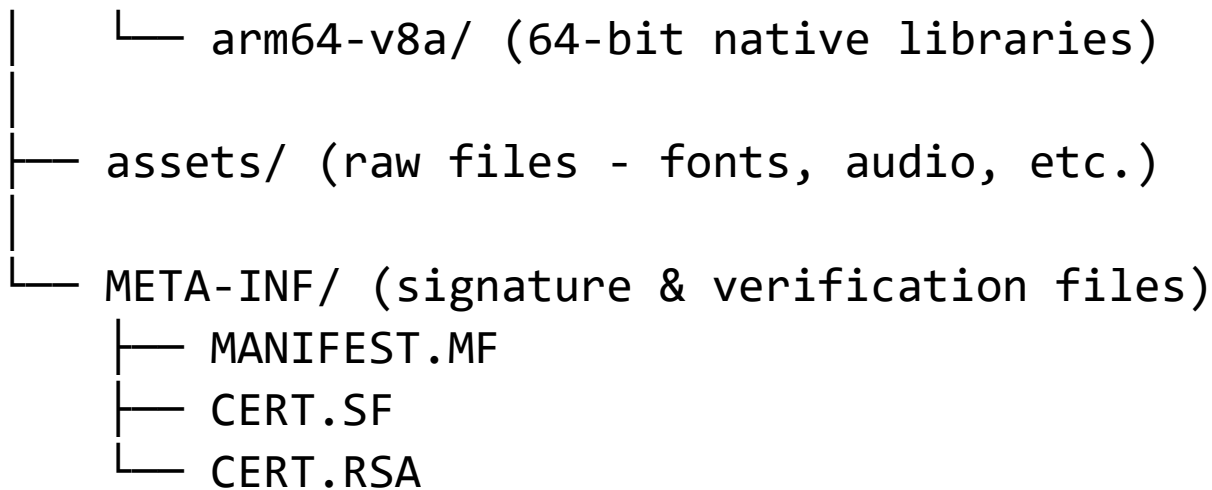
META-INF/ Folder

- App ke signature aur integrity verification ke liye use hota hai
- Ismein tino files hoti hain:
 - MANIFEST.MF – Files ki SHA-1 hashes
 - CERT.SF – Signed version of MANIFEST.MF
 - CERT.RSA – Public key certificate
- APK mod karne ke baad is folder ko resign karna zaroori hota hai

APK File Structure (Visual Overview)

app.apk (ZIP format mein)

```
|
|— AndroidManifest.xml (binary XML)
|— classes.dex (main bytecode)
|— classes2.dex (additional bytecode if multi-
dex)
|— resources.arsc (compiled resources)
|— res/
|   |— drawable/ (images)
|   |— layout/ (UI layouts)
|   |— values/ (strings, colors, dimensions)
|   |— mipmap/ (launcher icons)
|— lib/
|   |— armeabi-v7a/ (32-bit native libraries)
```



Important Note

- **AndroidManifest.xml** aur **classes.dex** reverse engineering ke liye sab se important files hain
- **META-INF** folder ko mod karne ke baad resign karna zaroori hai, warna app install nahi hogi
- **resources.arsc** mod karne ke liye special tools (jaise APKTool) ki zaroorat hoti hai

■ **Educational Purpose Only:** Yeh guide sirf educational aur learning purposes ke liye hai. APK reverse engineering ka istemal sirf apne own apps aur authorized environments mein karein. Kisi doosre ke app ko bina permission ke modify ya crack karna applicable cyber laws ke under punishable hai.

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